

Deploying GENERATIVE AI MODELS Efficiently

Enterprise deployment of Generative AI depends on the seamless optimisation of hardware and software, driving higher performance at lower cost. It highlights the purpose-built hardware powering GenAI and the software methods that help enterprises extract maximum efficiency.



Anish Kumar, AI Software Engineering Manager, Intel Technology India, during session at AI DevCon 2025 in Bengaluru

OpenAI's launch of ChatGPT powered by GPT-2 in mid-2020, showcased a model with 175 billion parameters, a monumental breakthrough at the time. By the arrival of GPT-4, parameter counts had surged into the trillions, enabling sophisticated chat assistants, code generation, and creative applications, yet imposing unprecedented strain on compute infrastructure.

Organisations are leveraging open source GenAI models, such as LLaMA, to streamline operations, enhance customer interactions, and empower developers. Choosing an LLM optimised for efficiency enables significant savings in inference hardware costs. The subsequent section explores how this is achieved.

As generative AI adoption soars, the significance of LLM parameters becomes clear

Since the public launch of ChatGPT, the adoption of generative AI has skyrocketed, capturing the imagination of consumers and enterprises alike. Its unprecedented accessibility empowered not just developers but also non-technical users to embed AI into their everyday workflows.

Central to this evolution is a fundamental measure of progress: LLM parameters, the trainable weights that are fine-tuned during learning to determine the model's capability. In 2017, early generative AI models based on the Transformer